From Simulated Falls to Freezing of Gait: A Controlled Study of Domain Shift in Wearable Accelerometry

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Question

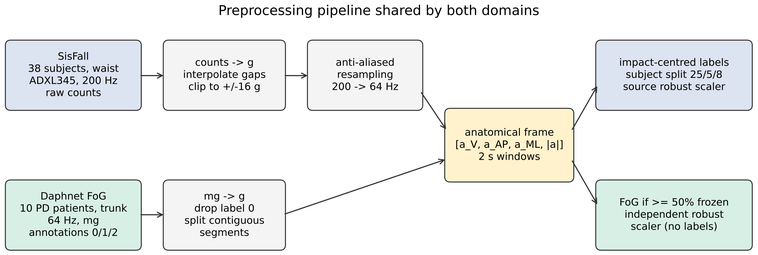
Wearable fall detectors are trained on healthy volunteers who simulate falls on mattresses. Patients with Parkinson's disease rarely fall that way: many falls start with **freezing of gait (FoG)**, a brief inability to step. We ask whether a deep fall detector trained on **SisFall** (38 healthy subjects) can flag FoG in **Daphnet** (10 Parkinson's patients) without any patient label, and if not, *why not*.

- **covariate shift**: normal movement differs (age, pathology);
- **concept shift**: the event to detect is a different phenomenon.

A controlled comparison

Sensor placement, units, labelling and training regime can all masquerade as "domain shift". Each is removed by construction:

- one accelerometer at the same site (waist / lower trunk), rotated into a vertical–antero-posterior–medio-lateral frame identified from the data;
- both domains in g at 64 Hz (SisFall downsampled with anti-aliasing);
- fall labels centred on the detected impact instead of the whole file;
- identical seeds, mini-batches, loss, optimiser and tuning budget for all four models.



What changes between the domains

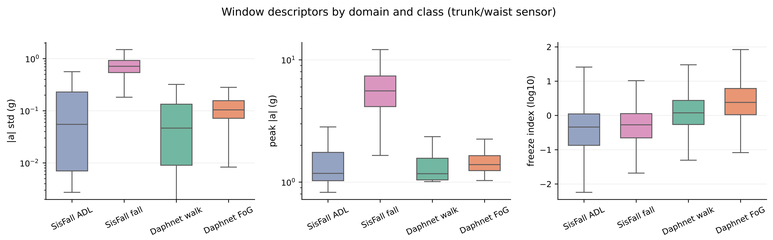


Figure 1:A fall window peaks at $5.6\,g$. A FoG window differs from Parkinsonian walking by only $0.2\,g$ in peak, twice the $|a|$ variability and more power in the 3–8 Hz freeze band.

Models

Model	Params	Selected configuration
CNN-1D	36k	3 conv blocks (32–64–128), global avg. pooling
LSTM	135k	2 layers, bidirectional, hidden 64
GRU	26k	2 layers, bidirectional, hidden 32
Transformer	70k	16 patches of 0.125 s, d=64, 2 layers, [CLS]

2 s windows of $[a_V, a_{AP}, a_{ML}, |a|]$, robust scaling, weighted BCE, AdamW, 3 seeds, equal tuning budget; threshold chosen on SisFall validation (F_2).

Main result

	SisFall test		Daphnet zero-shot	
Model	AUROC	false alarms/h	AUROC	FoG recall
CNN-1D	0.988	1.5	0.49	0.9%
LSTM	**0.999**	0.8	0.59	0.1%
GRU	**0.999**	**0.7**	**0.63**	0.0%
Transformer	0.998	0.9	0.62	0.0%
Freeze index (no training)	–	–	**0.66**	–

Near-perfect fall detectors become **blind to freezing**: they rank FoG barely above chance and stay silent at their validated threshold, while false alarms on Parkinsonian walking rise **15–18×**. Only the Transformer's patient-level 95% CI excludes chance (seed ensemble 0.65 [0.55, 0.73]).

In-domain vs. zero-shot

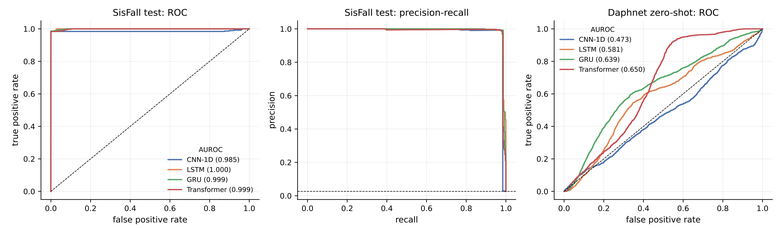
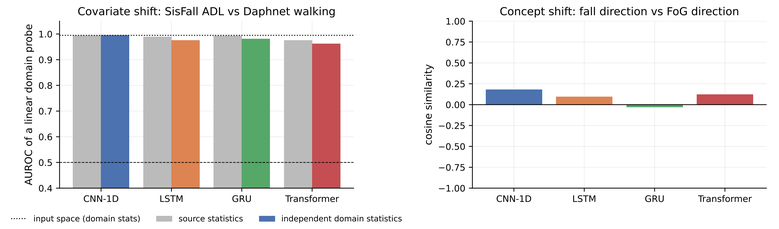


Figure 2:Seed-averaged ROC / PR curves. Left, centre: SisFall test subjects. Right: Daphnet.

References

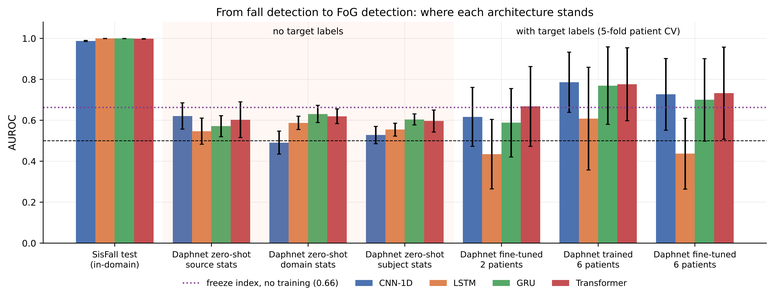
Sucerquia et al., SisFall, *Sensors* 2017 • Bächlin et al., Daphnet, *IEEE TITB* 2010 • Moore et al., freeze index, *J. Neurosci. Methods* 2008 • Nie et al., PatchTST, ICLR 2023 • Gulrajani & Lopez-Paz, DomainBed, ICLR 2021 • Ben-David et al., *Machine Learning* 2010 • Gretton et al., MMD, *JMLR* 2012.

Why transfer fails



- **Covariate shift persists.** A linear probe separates healthy ADL from Parkinsonian walking in every embedding (AUROC 0.96–0.995). Re-normalising Daphnet changes zero-shot AUROC by at most ±0.13, with no consistent gain.
- **Concept shift dominates.** In raw statistics, FoG and falls both mean "more movement" ($\cos = 0.77$). In the learned embeddings the fall direction is nearly orthogonal to the FoG direction ($\cos = -0.03$ to 0.18): the networks learned impacts, not agitation.
- **Negative transfer.** Fine-tuning from fall weights is worse than training from scratch for all four models with 6 labelled patients (AUROC -0.04 to -0.17).

Where each model stands



Take-aways

1. In-domain accuracy (AUROC ≈ 0.99 for every model) says nothing about transfer, and source validation cannot rank models by transferability.
2. With matched sensors and impact-centred labels, the detectors are specific to impacts, not to "abnormal movement": they stay silent on freezing.
3. The gap is a change of event, not only of population: physics-based features or event-agnostic self-supervised pre-training are better starting points than fall labels.
4. With 6 labelled patients a trunk sensor reaches AUROC ≈ 0.78 (CNN-1D, Transformer, GRU), the realistic ceiling for this setting.

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Code, notebooks and checkpoints: `parkinson-domain-shift` repository

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